

Pattern Block Exercises

Before starting, just play with the blocks for 5 minutes as a warm up and see what you can make.

Afterwards, split into groups of 2 or 3 and work on the problems.

Problem 1: Which of the shapes look the same when you flip them and which look like a mirror image?

Problem 2: Take an asymmetric shape and place it in the mirror so 3 copies of it appear (the original and 2 reflections). How many are the same and how many are mirror images?

Now close the mirror slowly until a fourth copy appears. How many are the same and how many are mirror images?

Now close the mirror until a fifth copy appears. How many are the same and how many are mirror images?

What happens with a 6th and 7th copy?

What do you think the pattern is? If we had 100 copies, or 101 copies, how many would be the same and how many would be mirror images?

Problem 3: Build designs with:

- One line of symmetry
- Rotational symmetry but no line of reflection symmetry
- No symmetry whatsoever

Problem 4: Take the asymmetric pink triangle. If you are allowed to:

- Rotate it by 90 degrees
- Flip it over its long edge

How many possible shapes can you make? Get lots of copies of the triangle and check.

Problem 5: Put the blocks in order from:

- Largest area to smallest area
- Largest perimeter to smallest perimeter

Problem 6: Can you make:

- Two different shapes with the same area?
- Same area, different perimeter?
- Same perimeter, different area?
- Same perimeter and area but different shapes?

Problem 7 (if time permits): Zendo. James to explain.